

HDOC DAY-39

C Programming

Name	Dhruv Pandey
SAP ID	590037660
Batch	15

Program 1: Variadic Function for Weighted Average

C Program

```
#include <stdio.h>
#include <stdarg.h>

double weightedAverage(int count, ...)
{
    va_list args;
    double value, weight;
    double weightedSum = 0.0, totalWeight = 0.0;
    va_start(args, count);
    for (int i = 0; i < count; i++)
    {
        value = va_arg(args, double);
        weight = va_arg(args, double);
        if (weight < 0)
        {
            printf("Error: Weight cannot be negative.\n");
            va_end(args);
            return -1;
        }
        weightedSum += value * weight;
        totalWeight += weight;
    }
    va_end(args);
    if (totalWeight == 0)
    {
        printf("Error: Total weight cannot be zero.\n");
        return -1;
    }
    return weightedSum / totalWeight;
}

int main()
{
    double result = weightedAverage(3, 80.0, 0.3,
                                    75.0, 0.2,
                                    90.0, 0.5);

    if (result != -1)
        printf("Weighted Average = %.2f\n", result);
    return 0;
}
```

Execution Screenshot - Program 1

```
dhruvpandey@DHRUVs-MacBook-Air ~ % nano day39.c
dhruvpandey@DHRUVs-MacBook-Air ~ % clang day39.c -oday39.c
dhruvpandey@DHRUVs-MacBook-Air ~ % ./day39
zsh: no such file or directory: ./day39
dhruvpandey@DHRUVs-MacBook-Air ~ % ./day39.c
Weighted Average = 84.00
dhruvpandey@DHRUVs-MacBook-Air ~ %
```

Program 2: Command-Line Integer Processing

C Program

```
#include <stdio.h>
#include <stdlib.h>

int isInteger(char str[])
{
    int i = 0;
    if (str[0] == '-' || str[0] == '+') i = 1;
    if (str[i] == '\0') return 0;
    while (str[i] != '\0')
    {
        if (str[i] < '0' || str[i] > '9') return 0;
        i++;
    }
    return 1;
}

void sortArray(int a[], int n)
{
    int temp;
    for (int i = 0; i < n - 1; i++)
        for (int j = 0; j < n - i - 1; j++)
            if (a[j] > a[j + 1])
                { temp=a[j]; a[j]=a[j+1]; a[j+1]=temp; }
}

double findAverage(int a[], int n)
{
    double sum=0;
    for (int i=0;i<n;i++) sum += a[i];
    return sum/n;
}

double findMedian(int a[], int n)
{
    if (n%2==0) return (a[n/2-1]+a[n/2])/2.0;
    return a[n/2];
}

int main(int argc, char *argv[])
{
    int numbers[100], n=0;
    for (int i=1;i<argc;i++)
    {
        if (isInteger(argv[i])) numbers[n++]=atoi(argv[i]);
        else printf("Invalid argument ignored: %s\n", argv[i]);
    }
    if (n==0) { printf("No valid integers found.\n"); return 0; }
    sortArray(numbers,n);
    printf("Valid integers in ascending order: ");
    for (int i=0;i<n;i++) printf("%d ",numbers[i]);
    printf("\nMinimum: %d\nMaximum: %d\n",numbers[0],numbers[n-1]);
    printf("Average: %.2f\n",findAverage(numbers,n));
    printf("Median: %.2f\n",findMedian(numbers,n));
    int found=0, secondLargest=0;
    for (int i=n-2;i>=0;i--)
        if (numbers[i] != numbers[n-1])
            { secondLargest=numbers[i]; found=1; break; }
    if (found) printf("Second-largest distinct value: %d\n",secondLargest);
    else printf("No second-largest distinct value.\n");
    return 0;
}
```

Execution Screenshot - Program 2

```
dhruvpandey@DHRUVs-MacBook-Air ~ % ./day39.c 12 5 abc 8 12 20
Invalid argument ignored: abc
Valid integers in ascending order: 5 8 12 12 20
Minimum: 5
Maximum: 20
Average: 11.40
Median: 12.00
Second-largest distinct value: 12
dhruvpandey@DHRUVs-MacBook-Air ~ %
```